Early-Warning Predictive Maintenance Using a Baseline Isolation Forest and a Multi-Stage Cascade Anomaly Detection Design

Danty D. Cook

Western Governors University

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A. Proposal Overview

This proposal addresses the need for earlier and more accurate identification of abnormal equipment activity. This project will examine data from industrial pump sensors to see if abnormal operating conditions can be identified before complete equipment failure. By examining the collection, processing, and analysis of this data, the project will examine if a multi-stage approach provides more valuable early warning alerts than a single-stage approach in a baseline model. The results of this project will aid in the maintenance process by improving the identification, prioritization, and review of potential issues before they cause unplanned downtime.

This need is especially important in the context that maintenance decisions are normally made after equipment has started to fail or after a failure has become more noticeable. In these instances, the absence of early identification of important warning signs can increase downtime, repair urgency, and operational significance. Because of this, this project is relevant in the context that it applies to the use of pump sensor data to not only identify unusual activity but also to provide early warning alerts that are significant enough to provide important maintenance information.

The goal of this project is to see if a multi-stage anomaly detection cascade architecture can improve alert accuracy over a single-model baseline by reducing false positives, quieting noisy alert rates, preserving useful anomaly sensitivity, and providing more meaningful results.

A.1 Research Question or Organizational Need

The organizational need that this project addresses is the need for earlier and more reliable detection of abnormal equipment behavior before a complete failure happens. In a predictive maintenance scenario, waiting until a problem becomes apparent can result in unnecessary downtime, rushed maintenance, and inefficient use of maintenance resources. A more robust

early warning system can help detect problems earlier and inform more informed maintenance decisions.

This project is guided by the following research question:

> Does a multi-stage anomaly detection cascade improve alert quality compared with a single Isolation Forest baseline for industrial pump sensor data?

More specifically, this project seeks to determine whether the multi-stage detection cascade can improve the detection of false positives, decrease noisy alert rate, maintain useful sensitivity to anomalies, and provide alerts that are more relevant to a real-world maintenance scenario.

The objective of this project is not only to detect unusual activity, but to determine whether a multi-stage detection process offers a more clear and useful early warning system than a single-stage broad anomaly detection model.

A.2 Context and Background

Industrial equipment can generate large amounts of sensor telemetry that may contain early signs of developing problems before a clear failure state occurs. Even so, detecting those warning signs is not always straightforward. Sensor data can be noisy, incomplete, or highly variable, which makes it difficult to separate meaningful degradation from short-lived fluctuations or unstable readings.

That challenge is what this project is built around. A single broad anomaly model can be useful for identifying unusual behavior, but it can also flag temporary spikes, harmless operating shifts, or sensor instability. In a maintenance setting, too many weak alerts can reduce trust in the output and make it harder to know when action is actually needed.

To address that problem, this project compares two model structures using industrial pump sensor data. The baseline model is a single Isolation Forest used as a broad anomaly detector. The comparative model is a three-stage cascade. In this design, the first stage uses a high-sensitivity Isolation Forest to screen broadly for potentially abnormal periods. The second stage uses a narrower Isolation Forest with stricter refinement and a reduced feature set. The third stage uses rule, profile, and historical confirmation logic to determine whether the remaining alerts are supported by patterns that are more consistent with meaningful abnormal behavior.

This comparison is meant to evaluate practical alert quality rather than simply maximize anomaly detection volume. The project is centered on whether a staged screening process can improve the usefulness of alerts for maintenance decision-making.

A.3 and A3A Summary of Published Works and Their Relation to the Project

Review of Work 1

Fast Anomaly Detection for Vision-Based Industrial Inspection Using Cascades of Null Subspace PCA Detectors (PMC12349016) (Zhou et al., 2024)

This published study investigates how anomaly detection can be made more effective in industrial inspection settings where rapid and accurate identification of defects is required. The authors observe that while many existing deep learning approaches have been successful in achieving high accuracy, they are often associated with computationally intensive models that are not very useful in practical scenarios. To overcome this limitation, the authors introduce a light-weight anomaly detection solution that leverages the strengths of a compact backbone model and a multi-stage cascade of detectors based on null-space principal component analysis (PCA). The proposed solution is more sensitive to anomalies by targeting the low-variance subspace of normal data, thus eliminating the need for computationally intensive reconstruction-based or distance-based approaches.

One of the major strengths of this study is the introduction of a cascade architecture that filters out obvious normal cases in the early stages and applies increasingly specific checks in the latter

stages. This approach helps to minimize false positives and computational requirements while still maintaining high levels of anomaly detection accuracy. The authors have demonstrated the effectiveness of the proposed cascade architecture on standard industrial datasets, which outperforms or matches the performance of more computationally intensive CNN-based models while being much more efficient.

Relation to This Project:

This study supports the relevance of comparing a single broad anomaly detection model with a multi-stage cascade of detectors. The results validate the idea that a cascade can help reduce unnecessary alerts, improve screening accuracy, and operate efficiently in industrial environments with limited resources. This aligns directly with the project's goal of evaluating whether a three-stage cascade built from Isolation Forest models and rule-based confirmation can generate more useful early warning alerts than a single unsupervised baseline model.

Review of Work 2

Adaptable and Unsupervised TinyML Anomaly Detection System for Extreme Industrial Environments (PMID 36850940) (Gómez et al., 2023)

This research proposes an unlabeled, unsupervised anomaly detection system specifically for extreme industrial environments, where typical cloud-based systems are inefficient due to connectivity issues and harsh environmental conditions. The authors highlight that typical industrial monitoring systems currently depend on external servers for data analysis from sensors, which introduces complexity and risks associated with network connectivity issues. To address these issues, the authors develop an end-to-end system that operates fully at the edge of the network using a low-power Internet of Things (IoT) device based on TinyML.

At the heart of this system, the authors develop an Isolation Forest model that is trained directly on the IoT device (an ESP32 microcontroller) to identify anomalies in real time. The system can be trained in as little as 1.2–6.4 seconds and make inference decisions in under 16 ms with a low RAM footprint, making it suitable for resource-limited industrial equipment. In addition, the

study incorporates blockchain technology to provide transparent and immutable logging of detected anomalies.

Relation to This Project:

This reference is highly relevant to the project's focus on unsupervised anomaly detection for industrial assets, particularly in scenarios where computational resources and connectivity are limited. It supports the use of the Isolation Forest model as a baseline anomaly detector and reinforces the value of real-time anomaly processing at the edge for pump sensor data. The emphasis on lightweight models and rapid classification provides practical context for building efficient unsupervised detectors and comparing them with a cascade architecture in this capstone project.

Review of Work 3

Cascading Multi-Agent Anomaly Detection in Surveillance Systems via Vision-Language Models and Embedding-Based Classification (arXiv 2601.06204v3) (Bose et al., 2025)

This study presents a multi-level cascade system for anomaly detection in surveillance videos that combines multiple detection paradigms to support both real-time performance and semantic interpretability. The authors highlight limitations of existing approaches: object detectors are fast but limited in semantic understanding, reconstruction-based methods lack contextual awareness, and large vision-language models provide rich semantic insight but at a high computational cost. To address these trade-offs, the paper introduces a cascading framework in which lightweight early stages—such as fast object detection and autoencoder-based reconstruction scoring—filter routine frames, while only uncertain cases are escalated to a higher-level semantic reasoning stage powered by a vision-language model. The system operates within a multi-agent architecture featuring adaptive escalation thresholds and a publish–subscribe backbone, enabling scalable deployment and asynchronous communication across modules. Experiments on the UCF-Crime dataset show that the cascade reduces average latency by approximately threefold compared to direct vision-language inference while maintaining strong detection accuracy and interpretable anomaly labeling.

Relation to This Project:

This study reinforces the foundational concept of cascading detection architectures by demonstrating how early-exit stages can reduce computational load and improve responsiveness without compromising accuracy or interpretability. Its multi-stage design closely parallels the project's cascade structure (broad anomaly detector → refined detector → confirmatory logic), illustrating how layered screening can enhance both efficiency and alert quality. Additionally, the inclusion of a higher-level reasoning stage highlights the broader value of escalation mechanisms for interpreting ambiguous conditions, aligning with the capstone's emphasis on producing meaningful early-warning alerts rather than excessive or noisy signals.

A.4 Summary of Data Analytics Solution

This project aims to provide a data analytics solution for early warning anomaly detection based on industrial pump sensor data. This solution is broken down into a Medallion Architecture consisting of Bronze, Silver, and Gold levels. This framework is useful in maintaining a streamlined workflow by breaking down data ingestion and standardization, analytical preparation, and model comparison into separate levels.

In the Bronze level, raw data ingestion and standardization are performed. In the Silver level, data cleaning, profiling, and analysis through exploratory data analysis and feature behavior analysis are conducted. In the Gold level, prepared data is used to compare a single Isolation Forest baseline model to a three-stage cascade comparative model.

The baseline model offers a single, broad-scope anomaly screening strategy for the pump sensor data. The cascade model is structured as a multi-level alert screening system. Stage 1 uses a highly sensitive Isolation Forest for broad screening. Stage 2 uses a more focused Isolation Forest for more refined screening with a smaller feature set. Stage 3 uses rule, profile, and historical validation logic to determine whether any remaining alerts are valid in terms of meaningful behavior patterns. This final stage is focused on profile violations in terms of normal

behavior patterns, short-term persistence checks, historical drift or shift checks, and multi-sensor validation.

The final project deliverables include notebook-based work at the Bronze, Silver, and Gold levels, prepared data, baseline and cascade model results, evaluation results, and supporting graphics. These elements are to be used to determine whether the three-stage cascade model is an improvement over the single-model baseline.

A.5 Benefits and Support of Decision-Making Process

The primary advantage of this project is the enhanced early warning system for abnormal equipment behavior before the occurrence of a complete failure becomes visible. In a predictive maintenance environment, this can aid in lowering the level of reactive decision-making and enabling earlier, more controlled actions to be taken.

A single, broad anomaly detector can detect abnormal behavior, but it can also produce alerts that are either too noisy or too frequent to be of much use on their own. The cascade architecture is designed to address this issue by including further refinement and confirmation steps for broad alerts. If this works, it can aid in lowering false positives, decreasing unnecessary alert volume, and providing alerts that are more believable and easier to act upon.

This means that the proposed solution can enable decision-making by enabling maintenance personnel to direct their attention to more significant abnormal periods, prioritize the examination of higher-risk abnormal behavior, and make better use of maintenance time and resources.

B. Data Analytics Project Plan

B.1 Goals, Objectives, and Deliverables

- **Goal 1**: Determine whether a three-stage anomaly detection cascade improves alert quality compared with a single Isolation Forest baseline for industrial pump sensor data.
  - **Objective 1.1**: Complete the Bronze layer of the Medallion Architecture by ingesting and standardizing the raw pump sensor data so it is organized in a consistent format for later analysis.
    - **Deliverable 1.1.1**: A Bronze notebook that ingests, validates the structure of, and standardizes the raw pump sensor data.
    - **Deliverable 1.1.2**: A standardized Bronze-layer dataset or saved output that is ready for downstream analytical use.
  - **Objective 1.2**: Complete the Silver layer of the Medallion Architecture by preparing, profiling, and analyzing the pump sensor data through pre-EDA and EDA workflows.
    - **Deliverable 1.2.1**: A Silver Pre-EDA notebook that performs initial cleaning, missing-value review, structural preparation, and readiness checks for deeper analysis.
    - **Deliverable 1.2.2**: A Silver EDA notebook that documents data profiling, exploratory analysis, and feature behavior analysis across normal and abnormal operating patterns.
    - **Deliverable 1.2.3**: Supporting Silver-layer outputs such as charts, summary tables, feature behavior findings, and analysis artifacts that help justify the later model design.
  - **Objective 1.3**: Complete the Gold layer of the Medallion Architecture by comparing the single Isolation Forest baseline with the three-stage cascade comparative model.
    - **Deliverable 1.3.1**: A Gold Preprocessing notebook that performs the preprocessing actions on the dataset to ready the data for machine learning model training.

        - **Deliverable 1.3.2**: A Gold modeling notebook section that builds and evaluates the single Isolation Forest baseline as the reference model for broad anomaly screening.
        - **Deliverable 1.3.3**: A Gold modeling notebook section that builds and evaluates the three-stage cascade comparative model, including:
            - **Stage 1**: Broad Isolation Forest for high-sensitivity screening
            - **Stage 2**: Narrow Isolation Forest for stricter refinement using a reduced feature set
            - **Stage 3**: Rule, profile, and historical confirmation layer
        - **Deliverable 1.3.4**: A Comparision notebook that loads the two models output date and performs a comparison with visuals showing how the baseline and cascade differ in alert behavior.
    - **Objective 1.4**: Evaluate whether the cascade improves practical alert quality compared with the baseline using alert-focused behavior.
        - **Delieverable 1.4.1**: A comparison of total alerts generated, false positives during normal periods, anomaly or failure-window capture, and reduction in alert volume from baseline to cascade.
        - **Deliverable 1.4.2**: A written interpretation explaining whether the cascade reduces false positives, reduces noisy alert volume, preserves useful anomaly sensitivity, and produces more operationally meaningful alerts.

B.2 Scope of Project

B.2.A Included in Project Scope

This project will involve the creation of a notebook-based data analytics process for industrial pump sensor data using a Medallion Architecture with Bronze, Silver, and Gold levels.

The Bronze level will involve data ingestion, validation, and standardization of the raw data from the industrial pump sensors for use in subsequent steps. The Silver level will involve data cleaning, examination for missing data, profiling, exploratory data analysis, feature behavior

analysis, and the generation of supplementary charts, summary tables, and analysis artifacts. The Gold level will involve model development and comparison between a single Isolation Forest model and a three-stage cascade comparative model.

The project will also involve the assessment of alert quality using alert-centric metrics, such as the total number of alerts produced, the number of false positives during normal times, the capture of anomalies or failure periods, precision and recall as applicable, and the reduction in the number of alerts from the baseline to the cascade model. The final project deliverables will include the Bronze, Silver Pre-EDA, Silver EDA, and Gold notebooks, as well as supplementary visualizations, comparison results, and a written explanation of the results.

A lightweight pipeline-ready structure, shared configuration, or modularity can also be employed as appropriate, provided that it aids in the data analytics process and does not increase the scope of the project beyond its current capstone project boundaries.

B.2.B Not included in Project Scope

The project does not involve full production deployment, a full streaming solution, extensive platform engineering, or multi-dataset automation as its main goals.

It also does not involve extensive production implementation of Kafka, PostgreSQL, or a production dashboard as its main goals. Although the overall project context may involve initial framework or plans for future pipeline, streaming, database, dashboard, or synthetic data generation, these are considered future work unless a small part of it is trivial to add without affecting the overall scope of the capstone.

The project is confined to the Data Analytics capstone scope, which means that the main focus is still on data preparation, exploratory analysis, feature behavior assessment, comparison of the anomaly model, and assessment of alert quality.

B.3 Standard Methodology

This particular project adopts a Medallion Architecture framework that is divided into Bronze, Silver, and Gold phases. This particular framework is adopted to ensure that data ingestion, analysis preparation, and final model comparison are divided into distinct phases. Adopting this framework ensures that the workflow is maintained in an organized manner and that there is a logical progression from raw pump sensor data to an evidence-based comparison of the two anomaly screening methods.

The Bronze phase is primarily concerned with data ingestion and standardization. During this phase, the raw industrial pump sensor data is ingested, assessed, and standardized into a consistent format that can be easily utilized during the analysis phase. The main objective of the Bronze phase is to ensure that there is a clean and organized starting point for the entire project.

The Silver phase is primarily concerned with data cleaning, profiling, exploratory data analysis, and feature behavior analysis. During this phase, the data is assessed for potential issues such as missing data, erratic data, and general data quality issues. The Silver phase is also utilized to analyze normal and abnormal operating conditions, assess sensor activity over time, and generate supporting charts, tables, and analysis artifacts that help to explain the behavior of the data prior to modeling.

The Gold layer is concerned with model comparison. In this phase, the prepared data is employed to compare the single Isolation Forest baseline model with the three-stage cascade comparative model. The baseline model is used as the reference model for general anomaly screening. The cascade employs a layered approach consisting of a general Isolation Forest, a more specific Isolation Forest, and a final rule, profile, and historical confirmation layer. The aim of the Gold layer is to ascertain whether the cascade enhances the quality of alerts by suppressing false positives, decreasing the amount of noisy alerts, maintaining useful anomaly sensitivity, and generating more meaningful alerts.

This approach is relevant to the project because it aligns with the scope of a Data Analytics capstone and offers a useful framework for transitioning from raw sensor data to a well-organized analytical comparison. It aligns with the notebook-based approach utilized in the project and ensures that the analysis is centered on data preparation, feature behavior analysis, anomaly model comparison, and analysis of results instead of production deployment.

B.4 Timeline and Milestones

Milestone or deliverable	Duration (hours or days)	Projected start date	Anticipated end date
Project Kickoff, Research, Topic Decision, & Dataset Acquisition	11 Days	*2025-12-01*	*2025-12-12*
Bronze Layer – Raw Data Ingestion & Standardization (Deliverables 1.1.1 & 1.1.2)	7 Days	*2025-12-13*	*2025-12-20*
Brone Validation & Metadata Structuring	3 Days	*2025-12-27*	*2026-01-02*
Silver Pre-EDA – Cleaning, Structuring Checks, Nulls/Duplicates Review (Deliverable 1.2.1)	14 Days	*2026-01-03*	*2026-01-16*
Silver Artifacts – Feature Profiles, Sensor Behavior,	4 Days	*2026-01-17*	*2026-01-20*

Summary Tables (Deliverable 1.2.3)			
Gold Preprocessing – Feature Engineering, Scaling, Alignment (Deliverable 1.3.1)	7 Days	*2026-01-21*	*2026-01-27*
Gold Baseline Modeling – Single Isolation Forest (Deliverable 1.3.2)	5 Days	*2026-01-28*	*2026-02-01*
Gold Cascade Modeling – Stage 1 (Broad IF)	3 Days	*2026-02-02*	*2026-02-04*
Gold Cascade Modeling – Stage 2 (Narrow IF)	3 Days	*2026-02-05*	*2026-02-07*
Gold Cascade Modeling – Stage 3 (Rule/Profile Confirmation)	4 Days	*2026-02-08*	*2026-02-11*
Model Evaluation – Alert Counts, Anomaly Capture (Deliverable 1.4.1)	7 Days	*2026-02-12*	*2026-02-18*
Model Comparison Notebook – Visuals & Side-By-Side Behaviors	4 Days	*2026-02-19*	*2026-02-22*
Written Interpretation – Alert Quality	3 Days	*2026-02-23*	*2026-02-25*

Analysis (Deliverable 1.4.2)			
Final Completion, Proofreading, Formatting (Task 2/Task 3 Assembly)	2 Days	*2026-02-26*	*2026-02-28*
Capstone Completion and Submission	---	*2026-02-28*	*2026-02-28*

B.5 Resources and Costs

The resources required for this project include hardware, software, and analyst work time. Because this project is being completed as a capstone-scale analytical and experimental environment, the overall direct cost is limited. The estimates below reflect the resources needed to complete the current notebook-based analysis, anomaly model comparison, and written documentation. They are not intended to represent the cost of a full production deployment.

- **Existing computer or laptop**: $0 additional cost
  An existing personal system is being used to complete the project's notebook development, data analysis, model testing, and documentation.
- **Python development environment**: $0 additional cost
  An existing Python environment is being used for data preparation, anomaly modeling, and evaluation.
- **JupyterLab**: $0 additional cost
  JupyterLab is being used as the main notebook environment for the Bronze, Silver, and Gold analytical workflow.
- **Python libraries (pandas, NumPy, scikit-learn, matplotlib)**: $0 additional cost
  These open-source tools are used for data handling, preprocessing, anomaly detection, and visual communication of results.

- **Docker and WSL Ubuntu**: $0 additional cost
  These tools are used to maintain a consistent development and testing environment and to support lightweight containerization of the project environment.
- **GitHub**: $0 additional cost
  GitHub is used for version control and organization of notebooks, scripts, configuration files, and related project materials.
- **Weights & Biases**: $0 additional cost within the current scope
  Weights & Biases is used for artifact handling and experiment support, including organization of analysis outputs and related files.
- **Estimated labor**: 65 hours × $25 per hour = $1,650
  This estimate covers the time required for data ingestion, cleaning, exploratory analysis, feature behavior review, model development, alert-quality evaluation, visualization, and final written documentation.

Because this project is not a production-grade implementation, these estimates do not include enterprise-scale infrastructure, hosted services, streaming platforms, database resources, dashboard platforms, monitoring systems, or ongoing operational support. Those larger-scale costs fall outside the scope of this capstone and would apply only in a future production environment. For the current project, the primary cost is the analyst time required to complete the work.

B.6 Criteria for Success

The project will be deemed successful if it achieves a complete and functional analytical comparison between the single Isolation Forest baseline and the three-stage cascade comparative model.

The project will be deemed successful if it achieves the successful completion of the major project components. This includes the completion of the Bronze notebook for ingestion and standardization, completion of the Silver Pre-EDA and Silver EDA notebooks for cleaning,

profiling, exploratory data analysis, and feature behavior analysis, and completion of the Gold notebook for baseline and cascade model comparison.

The project will also be deemed successful if it achieves a clear comparison of alert quality based on the proposed evaluation criteria. This includes demonstrating the total number of alerts produced by each method, the identification of false positives during normal times, analysis of anomaly or failure window capture, and the reduction in alert volume from the baseline to the cascade. Precision and recall can also be used if applicable.

The project will not be considered unsuccessful if it does not achieve perfect predictive accuracy. Instead, the project will be considered successful if it is completed as planned and if the analysis is able to clearly conclude whether the cascade improves alert quality by reducing false positives, reducing the volume of noisy alerts, preserving useful anomaly sensitivity, and producing more meaningful alerts than the baseline.

C. Design of Data Analytics Solution

C.1 Hypothesis

The hypothesis for this project is that the three-stage anomaly detection cascade will result in higher quality alerts than the single Isolation Forest baseline when used on industrial pump sensor data.

More specifically, the hypothesis is that the three-stage cascade will result in a reduction in false positives, a reduction in the number of noisy alerts, a retention of useful sensitivity to anomalies, and alerts that are more meaningful than those produced by the baseline alone. The goal of this comparison is not simply to see which model identifies more anomalies, but to see if the multi-stage alerting process results in a more useful alerting outcome.

C.2 Analytical Method

This project employs anomaly detection as its main analytical technique because the aim is to find anomalies in the operating data of industrial pump sensors and compare the relative usefulness of two alert screening strategies.

The baseline model consists of a single Isolation Forest. This model is the baseline strategy for general anomaly screening of the preprocessed pump sensor data. It is used to determine how a single-model anomaly detector performs when directly applied as a screening strategy.

The comparative model consists of a three-stage cascade:

- Stage 1: a general Isolation Forest for highly sensitive screening
- Stage 2: a more selective Isolation Forest for more refined screening with a smaller feature set
- Stage 3: a rule, profile, and historical verification layer

The aim of the cascade is to determine if staged screening is more effective at producing high-quality alerts than a single general model. The first stage is meant to cast a broad net and screen for potentially anomalous activity. The second stage is meant to refine this finding by applying a more stringent filter over a smaller feature set. The third stage is meant to verify whether the remaining alerts are consistent with significant patterns of meaningful behavior rather than fleeting anomalies or isolated instances of instability.

In the final verification stage, alerts are assessed for:

- profile violations against normal operating patterns
- persistence against short rolling windows
- historical drift or shift
- multi-sensor verification

This final stage is meant to assist in distinguishing more significant alert candidates from less significant anomaly signals that may not be indicative of useful maintenance issues.

C.2.A Justification of Analytical Method

These methods are appropriate because the project is focused on the application of anomaly screening as opposed to just anomaly detection volume. A single Isolation Forest baseline is a good reference model for general anomaly detection in multivariate sensor data. It is appropriate because it provides a simple means of determining anomalous behavior without adding unnecessary complexity to the baseline comparison.

On the other hand, a single general detector can also generate weak or noisy alerts. In the context of predictive maintenance, this can make the results of the output less useful if too many alerts are generated during normal operation or if transient behavior is identified as significant events. Because of this, the three-stage cascade comparative model is employed to determine if a process improvement can be gained through a layered approach.

Each of the stages in the cascade has a particular use. The first stage is used for broad screening to retain useful sensitivity. The second stage uses more aggressive refinement so that the initial anomaly pool can be reduced. The third stage uses confirmation logic based on profile behavior, persistence, drift, and corroboration so that alerts can be assessed relative to more interpretable evidence. This is directly relevant to the project's research question because it tests whether the cascade can reduce false positives and noisy alert volume while retaining useful anomaly sensitivity.

The methods are also appropriate because they are relevant to the scope of a Data Analytics capstone project. The project is focused on data preparation, feature analysis, model output comparison, and interpretation of alert quality as opposed to developing a production-grade system..

C.3 Tools and Environments

The approach for this project is being implemented mainly in Python with a JupyterLab environment. JupyterLab is utilized for the main implementation because it allows for exploratory analysis, data preparation, feature examination, and model development in a manner that is easily documented and assessed. This is an excellent match for a Data Analytics capstone project because the project requires the analysis of sensor data, evaluation of preprocessing strategies, and comparison of model results in a manner that is easily documented and assessed.

The project utilizes pandas and NumPy for data manipulation and preparation. These libraries are utilized for loading the industrial pump sensor data, structuring the data, preparing and cleaning the data values, examining feature behavior, and preparing data for modeling. scikit-learn is utilized for the anomaly detection modeling, including the single Isolation Forest model and the Isolation Forest models in the cascade. matplotlib is utilized for visualization, including time series plots, anomaly plots, and comparison plots that can help illustrate how the single model and the cascade compare across the data.

The current approach is notebook-driven, but the project has been structured in a light pipeline-ready fashion wherever possible. The project is being developed within Docker and Ubuntu using WSL, which assists with maintaining a consistent environment and is conducive to reproducibility. The project is also being maintained within GitHub for version control purposes so that the notebooks, scripts, configuration files, and other related assets can be tracked as the work progresses. Weights & Biases is also being used for artifact management and experiment tracking so that the generated analysis files and other related outputs can be kept organized.

While the capstone itself will continue to be notebook-driven for analytical purposes, the project structure is conducive to extension into a more scripted and modular approach if necessary. This will allow the current scope to remain realistic and submission-driven while still being able to keep things organized and reproducible where possible.

These tools and environments are relevant because they assist with the immediate analytical needs of the project without expanding the capstone into a full engineering or deployment

project. They offer all that is required to clean the data, profile the features, develop the baseline and cascade models, assess the quality of the alerts, and communicate the results.

C.4 and C.4.A Methods and Metrics to Evaluate Statistical Significance

This project will assess the difference between the single Isolation Forest baseline and the three-stage cascade comparative model using a statistical test and model evaluation metrics with a focus on alert quality.

Statistical Test Plan

The main hypothesis of this project is that the three-stage cascade is better at improving alert quality than the single Isolation Forest baseline. For the statistical assessment of the project, the hypothesis to be tested is that the cascade has a lower false positive rate during normal operating periods than the single Isolation Forest baseline.

- **Null hypothesis (H0):** There is no statistically significant difference between the false positive rate of the single Isolation Forest baseline and the false positive rate of the three-stage cascade during normal operating periods.
- **Alternative hypothesis (H1):** The three-stage cascade will result in a lower false positive rate during normal operating periods than the single Isolation Forest baseline.
- **Planned statistical test**: A paired comparison test will be conducted across equivalent evaluation periods. The specific paired test to be conducted will be determined by the distribution of the differences in false positive rates.
- **Metric generated from the test**: The chosen paired test will generate a test statistic and associated p-value that will be used to determine statistical significance.
- **Alpha value**: $\alpha = 0.05$

If the calculated p-value is less than 0.05, the null hypothesis will be rejected, and the project will conclude that there is sufficient evidence that the cascade is effective at reducing false positives compared to the baseline during normal operating periods.

Model Evaluation Plan

Model 1: Baseline Model

- **Type of model**: Unsupervised anomaly detection
- **Algorithm**: Single Isolation Forest
- **Performance metrics**:
  - total alerts generated
  - false positives during normal operating periods
  - anomaly or failure-window capture
  - precision and recall where practical
- **Benchmark for success**:
  The baseline model will be considered successful if it produces a usable set of anomaly alerts that can be evaluated across the full analysis period and provides a clear reference point for broad anomaly screening.

Model 2: Comparative Model

- **Type of model**: Multi-stage anomaly detection cascade
- **Algorithms / logic used:**
  - **Stage 1**: Broad Isolation Forest
  - **Stage 2**: Narrow Isolation Forest
  - **Stage 3**: Rule / profile / historical confirmation layer
- **Performance metrics**:
  - total alerts generated
  - false positives during normal operating periods
  - anomaly or failure-window capture
  - precision and recall where practical
  - reduction in alert volume from the baseline to the cascade
- **Benchmark for success**:
  The cascade model will be considered successful if it reduces false positives and reduces noisy alert volume compared with the baseline while still preserving useful anomaly

sensitivity, as shown by continued capture of meaningful anomaly or failure-related windows.

The results of both models will be compared using summary tables, visualizations, and alert-based analysis so that the meaningful difference in alert quality can be easily interpreted.

C.4.A. Justification of Methods and Metrics

These techniques and metrics are relevant because the project has a focus on the quality of alerts, rather than just the overall number of anomalies found. In a predictive maintenance context, a model is more valuable if it can improve the filtering out of unnecessary alerts while still detecting significant abnormal activity.

The statistical analysis is also included in order to assess whether the difference in false positive behavior between the baseline and the cascade is likely to represent a significant change rather than mere random fluctuation. False positives during normal operating periods are a relevant target for comparison because they directly measure the noise of alerts, which is one of the key problems that the cascade is intended to solve.

The metrics of model performance are also relevant because they measure the underlying purpose of the project. Total number of alerts produced and alert volume reduction can help to assess whether the cascade produces a more manageable screening outcome. False positives during normal operating periods can help to assess whether unnecessary alerts are reduced. Anomaly or failure window capture can help to assess whether significant anomaly sensitivity is preserved. Precision and recall, if possible, can provide further support if the available labels or proxy labels are sufficient to provide meaningful assessment.

These techniques collectively allow the project to assess both whether the cascade demonstrates a significant reduction in false positives and whether it produces a more significantly useful alerting outcome than the single Isolation Forest baseline.

C.5 Practical Significance

The relevance of this project to practice will depend on whether the final output is an improvement over the current screening and review of abnormal equipment behavior in a predictive maintenance context.

The project is relevant if the three-stage cascade is able to minimize unnecessary alerts while still being able to capture valuable abnormal behavior. This will ensure that the output of the project is more than a technical comparison of two model structures. It will ensure that the output is a more valuable screening tool that can help draw attention to more valuable abnormal periods and help prioritize maintenance review.

From a practical perspective, a more valuable screening tool will help ensure that less time is wasted responding to weak signals and help ensure that it is easier to determine when a developing problem is worth further review. This project is therefore important not because it will ensure perfect detection, but because it will determine whether a layered screening process for anomalies will provide more valuable alerts than a broad baseline.

This makes the project relevant to real-world decision-making in predictive maintenance, where the usefulness of an alert can be just as important as the ability to detect unusual behavior in the first place..

C.6 Visual Communication

The findings of this project will be communicated using visualizations that help show both sensor behavior and model output over time. These visual representations are important because they make it easier to interpret how the baseline and cascade behave and how the alerting patterns differ between the two approaches.

The project will use visual representations such as:

- time-series plots showing sensor behavior across the observed operating period

- anomaly overlays that show where the baseline model flags abnormal periods
- comparison visuals that show how cascade-confirmed alerts differ from the broader baseline alerts
- summary charts or tables that compare total alerts, false positives, and other alert-focused evaluation results

These visualizations will support the written analysis by showing whether the cascade narrows noisy alert behavior and produces a more focused result than the baseline alone. They also help connect the technical outputs of the model comparison to the practical goal of improving alert quality in a maintenance setting.

D. Description of Dataset

D.1 Source of Data

The dataset used for this project is the “Pump Sensor Data” dataset by “UnknownClass” that is publicly available on Kaggle.com.

D.2 Appropriateness of Dataset

The dataset is made up of multivariate time series data from sensor readings of an industrial water pump system, which provides the water for the living area that is not part of a larger town. The dataset was created using 52 sensors that were placed on different core components, systems, and points on and around the water pump over the course of a year. During the time the data was captured, a total of seven failures were recorded and marked in the dataset. This makes the dataset a suitable candidate for analysis related to anomaly detection and predictive maintenance. Additionally, with the failures and recovery window clearly marked, the dataset is a suitable candidate for potential identification of failure prevention early warning alerts.

D.3 Data Collection Methods

The data was collected as time-series telemetry from a monitored pump system sensors and then structured for use in analytics and machine learning. The compiled data was then hosted as publicly available dataset on Kaggle.com as a downloadable comma separated file (.csv), https://www.kaggle.com/datasets/nphantawee/pump-sensor-data/data.

D.4 Observations on Quality and Completeness of Data

The dataset is suitable for the project, but it still needs to be prepared before it can be modeled well. Just like most real-world sensor datasets, it could contain missing data, noise, transient peaks, and varying patterns among sensors. These are not uncommon issues in predictive maintenance data, but they do impact how the analysis needs to be done.

Instead of viewing these as drawbacks, they actually make the dataset more representative of the problem domain being analyzed. The dataset will be evaluated for its quality and completeness during the preprocessing stage, and any actions taken will be recorded as part of the analysis process.

D.5 and D.5.A Data Governance, Privacy, Security, Ethical, Legal, and Regulatory Compliances

- **Data governance**: Dataset fields, transformations, and feature definitions will be documented in a data dictionary and version-controlled to ensure traceability and reproducibility.
- **Privacy**: The main, pump sensor, dataset contains no direction personal identifying information. However, if a dataset that contains any human-related identifiers is utilized to demonstrate the ambiguity of the pipeline's functionality, it will be removed or anonymized before analysis.

- **Security**: Any data that does not fall under open source or public use will be stored locally in secure project folders with access restricted to the analyst account. Sensitive materials will not be shared publicly.
- **Ethical, legal, and regulatory compliance considerations**: The analysis will follow terms of use/licensing for the dataset. Outputs will be reported transparently, including any limitations, assumptions, and risks of false positives.

Several measures will be taken during the project. The origin and terms of use of the dataset will be well documented. The preprocessing steps and decisions made in the models will be well documented to enable the process to be traced and replicated. The results of the project will be presented in a way that the flags for anomalies are explained as warnings of high risk instead of failures. The project files and results will be maintained in a controlled environment.

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